Introduction to Software Engineering

Project Idea & Team Contract



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Project Idea & Team Contract

Objectives

This document focus on the following topics:

- ✔ Describing the project idea at a high level so the teaching staff can approve it
- ✔ Agreeing on how the team will work together (team contract)
- ✔ PA0 is not graded, but it is required before the team starts PA1

1 Team Information

Item	Value
Team ID	*T01*
Team name	*The Artisans*
Team lead	*Lại Phan Nhật Nam*
Communication channel	*Discord server*

Student ID	Full name	Email	Main role	Backup role
24127455	Lại Phan Nhật Nam	lpnnam2424@clc.fitus.edu.vn	Team lead	Designer
24127259	Phạm Hoàng Khôi Nguyên	phknguyen2419@clc.fitus.edu.vn	Business analyst	Tester
24127459	Trần Văn Nam	tvnam2419@clc.fitus.edu.vn	Designer	Developer
24127308	Nguyễn Hoàng Tiến	nhtien2435@clc.fitus.edu.vn	Developer	Business analyst
24127409	Ngô Phúc Khang	npkhang2416@clc.fitus.edu.vn	Tester	Team lead

2 Project Idea Overview

2.1 Project Name and One-Sentence Pitch

SplitNest - a web application that helps manage shared expenses.

2.2 Problem and Motivation

Shared expenses are typically managed in a messaging app or a spreadsheet. However, with those approaches, each update is inconvenient as it has to be done manually, which can cause inconsistent information between members.

A clear interface is needed to display up-to-date information and to confirm settlements: the payer marks a debt as paid, and the receiver confirms it, which updates the debt's status.

2.3 Target Users

User group	What they need from the system
Friends/ Family	Coherent data showing which expense is settled and which one is not.
Treasurer	Keep track of those who have not contributed to the fund yet.

2.4 Main Features

#	Feature	Short description	Must-have?
1	Account registration	Users can create an account and save their bank details (bank name, account number, holder name) as text.	Yes
2	Group creation and invitation	Users are able to create a group and send invites to other users to join the group. A quick join is also available through QR scanning..	Yes
3	Shared expense creation and user tagging	Users can create a shared expense inside a group, set the amount and split, and tag the members who must pay their share.	Yes
4	Debt watching	Users can see the debts they owe and are owed, mark a debt as paid, and confirm payments they receive.	Yes
5	VietQR Generation	From the user's saved bank details, generate a VietQR code for payment.	No
6	History and summary	Show expenses and settlement in a time-based order.	No

7	Debt reminder	When a debt remains unpaid after a period, both the lender and borrower will receive proper notification.	No
8	Chats and notes	Personal and group chatting, send notes regarding unsettled debt.	No

2.5 Similar Products

Splitwise: the difference being we opt for a more Vietnamese-first UI design, while also offering VND currency and VietQR support to fit the context of how payment usually works in Vietnam.

2.6 Platform and Technology Ideas

Supported Device: Web

Languages: HTML, CSS, JavaScript, Python

Frontend: React

Backend: Flask

Database: MongoDB Atlas

Deployment: Vercel

2.7 Feasibility, Scope and Open Questions

Feasibility: We believe the 4 must-have features can be built and deployed within weeks 8–10. The two developers own the backend and technical decisions. The designer, business analyst and tester implement frontend screens and automated tests under developer review. We simplify the features: bank details are stored as text (no real bank linking), settlement is "payer marks paid, receiver confirms", and invites use a join link or QR code. Because deployment is required in week 10, we deploy a minimal working app (frontend, Flask API, database) in week 8, then release every merged feature to it, and freeze new features at the start of week 10.

Out of scope: real bank integration or payment automation, multi-currency, native mobile app, email or push notifications, real-time chat.

Risks:

(1) Deployment problems (hosting limits, environment configuration, database access) found late,

we mitigate by deploying in week 8.

(2) Final bug fixes and demo preparation colliding in week 10.

Questions for the TA:

(1) Are we allowed to change the main features or purpose of the application in subsequent weeks?

(2) Do all features proposed in PA0 have to be implemented or can we discard some?

(3) Are automated tests expected, and at what coverage?

3 Team Contract

3.1 Goals and Expectations

The team aims for the highest grade possible, preferably 9 on a 10 scale. Every week, each member should spend at least 5 hours on the project doing their tasks or suggesting potential improvements that can be made in the project. During week 8-10, minimum working hours go up to 6 hours for final refinement.

3.2 Communication

- ✔ Discord channel, direct message or phone call
- ✔ Within 4 hours during day, 10 hours at night
- ✔ Github stores the repository.
- ✔ Jira acts as a task board.
- ✔ Google Drive is used as a documents archive.

3.3 Meetings

- ✔ Weekly status meeting with the TA: Saturday, 3:00 PM, online.
- ✔ Internal team meetings: Monday, 7:30 PM.
- ✔ Attendance rule:

  (1) Late attendance that exceeds 15 minutes without permission will affect final contribution points.
  (2) Inform the Team lead or backup Team lead for absence notice 12 hours in advance. For

urgent cases, at least a message is fine.

(3) During meetings, members need to disclose every work, question, suggestion they have.

✔ Weekly status report deadline chosen by the team: Sunday, 7:30 PM

3.4 Roles and Responsibilities

See the table in section 1. Extra:

(1) Lại Phan Nhật Nam sets up the repository and CI.

(2) Nguyễn Hoàng Tiến and Trần Văn Nam prepare presentation content.

(3) Phạm Hoàng Khôi Nguyên designs the slides.

(4) Ngô Phúc Khang keeps the bug list.

3.5 Work Standards

✔ Internal deadline: each member finishes their part 12 hours before the PA deadline.

✔ Quality: every document section and pull request is reviewed by at least one other member

✔ Tools: every task is on the task board with an owner and due date; code goes through pull requests

3.6 Use of AI Tools

Allowed tools: Claude, ChatGPT, Copilot. Allowed for: brainstorming, review, boilerplate, explaining errors. Not allowed: submitting AI output unread. Every output must be reviewed and edited by a member who can explain it. Trần Văn Nam keeps the AI usage log (rotate each PA if preferred).

3.7 Decision Making

We aim for consensus. If it is not reached in one meeting, we vote by majority and the team lead breaks ties. Technical decisions are made by the developers, requirement decisions by the business analyst.

3.8 Handling Conflicts and Missed Work

- ✔ Step 1: talk directly with the member and agree on a fix and a new date
- ✔ Step 2: discuss in a team meeting and record it in the weekly status report
- ✔ Step 3: inform the TA; the issue is taken into account in the individual contribution factor
- ✔ A task more than 2 days late without notice counts as missed work and moves to Step 2.

3.9 Signatures

Student ID	Full name	Signature	Date
24127455	Lại Phan Nhật Nam	NAM	12/10/2026
24127259	Phạm Hoàng Khôi Nguyên	NGUYÊN	12/10/2026
24127459	Trần Văn Nam	NAM	12/10/2026
24127308	Nguyễn Hoàng Tiến	TIẾN	12/10/2026
24127409	Ngô Phúc Khang	KHANG	12/10/2026

4 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Lại Phan Nhật Nam	Claude	Grammar checking. Checking feasibility for implementation.	Correct some unintended or ambiguous wording. Revise how the feature should behave from the agent's suggestion (change from transaction API which we have no access to confirmation between users).